Executable Evaluation Contracts for Target-Weight Trading Pipelines

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Abstract

A sequence of target weights is not a complete backtest specification. Reported performance also depends on when signals become executable, what residual cash earns, how turnover incurs cost, and whether the comparator bears the same exposure. We encode these choices as executable contracts in QUANTCORTEX, an open-source research and paper-trading system. The contracts cover causal inputs, bounded targets, adjusted-close pseudo-share execution, explicit cash and costs, separate attribution and tradable comparators, fail-closed order state, and provenance for every published result artifact.

We apply the protocol to a fixed six-ETF rotation strategy over 2,011 U.S. trading sessions from 2018–2025. Under the audited conventions, the strategy records a 1.40% net CAGR and a -0.15 conventional sample Sharpe relative to a short-Treasury ETF cash proxy (SHV). A causal target-exposure comparator, after its own modeled costs, records 5.72% and 0.62; the strategy's annualized arithmetic shortfall is -4.14% [-6.70%, -1.63%]. An exact daily identity decomposes its -0.90% annualized excess over cash into 3.90% from passive risky exposure, -3.38% from active risky allocation, 0.30% from exposure timing, and -1.72% from modeled cost. The active-allocation interval is [-5.85%, -0.90%]. Holding the targets and data fixed, omitting costs changes the Sharpe to 0.14; deliberately assigning a close-derived target to the return ending at that close changes it to 0.03.

This is an evaluation methodology and a negative case study, not a new alpha model. It shows that deterministic, artifact-traced computation can still produce an economically misleading estimate when timing, cash, costs, and comparator exposure are left implicit.

1 Introduction

A target-weight vector is a convenient interface between prediction, portfolio construction, and execution. Qlib and FinRL use modular research pipelines [Yang et al., 2020, Liu et al., 2020]; FinRL-X extends the interface through broker execution [Yang et al., 2026]. A target, however, is an intention rather than a realized return. A complete evaluation must also state when that target can be held, what uninvested capital earns, how turnover becomes cost, which holdings are credited with each return interval, and what happens when an order outcome is uncertain.

These choices correspond to separate hypotheses. The information set may be causal while the cash account is incomplete. The accounting may be correct while the code is nondeterministic. A reproducible backtest may still offer no investment value, and a valid simulation says little about whether retrying an uncertain broker request is safe. Treating these questions as one notion of "correctness" obscures the source of failure.

Preprint.

The distinction is particularly important in financial machine learning, where observations are dependent, effective samples are limited, and repeated strategy search creates substantial false-discovery risk [White, 2000, Harvey et al., 2016, Bailey and López de Prado, 2014, Bailey et al., 2017, Arnott et al., 2019]. We ask a narrower systems question: how can a target-weight pipeline make its economic and operational semantics executable, testable, and traceable from the information set to paper-order state?

The primary contribution is the evaluation methodology. The strategy is a test case, and the software is a reference implementation. Specifically, we (i) define a machine-readable evaluation contract and canonical target tape for information timing, weights, overlays, execution lag, cash, costs, comparison, and uncertain order outcomes; (ii) formalize an exact daily return decomposition for portfolios with changing risky exposure and preserve that identity under joint block resampling; and (iii) bind the fixed case-study evidence to its configuration, source tree, environment, input digest, and published artifacts. The empirical result is intentionally negative: it localizes why one fully traced rotation strategy fails rather than presenting it as profitable alpha.

2 Related work

Financial backtest validity. Sharpe-ratio inference depends on distributional and serial-dependence assumptions [Lo, 2002]. Data snooping, repeated trials, and researcher degrees of freedom can make a selected backtest appear stronger than the process that generated it [White, 2000, Bailey and López de Prado, 2014, Bailey et al., 2017, Harvey et al., 2016]. Arnott et al. [2019] organize these risks into a research protocol. In a recent preprint, Zhang et al. [2026] estimate decision-time leakage by changing one evaluation convention at a time across multiple models and panels. Our same-close diagnostic is narrower: it is one contract test inside a broader accounting and execution audit, not a general leakage benchmark.

Attribution and matched comparison. Performance evaluation has long separated investment decisions into interpretable return components [Fama, 1972, Brinson and Fachler, 1985]. We apply that principle to a portfolio whose risky exposure changes over time. The identity separates passive exposure, exposure timing around the sample mean, active allocation, and modeled cost. It is an exact arithmetic decomposition, not a new geometric linking method. The case-study signal draws on cross-sectional, time-series, and residual momentum ideas [Jegadeesh and Titman, 1993, Moskowitz et al., 2012, Blitz et al., 2011], but is used only as a fixed test vehicle.

Research software and implementation risk. Qlib and FinRL provide modular financial-AI workflows [Yang et al., 2020, Liu et al., 2020]; FinRL-X emphasizes a shared weight interface across research and deployment [Yang et al., 2026]. Computational reproducibility remains difficult even with code and data. Across 1,000 computational reproduction attempts in finance, the original code regenerated the reported result exactly only 52% of the time [Pérignon et al., 2024]. In a recent preprint, Yin et al. [2026] compare five engines across 15 strategies and find material divergence from implementation choices, including transaction costs. Our two-engine result is deliberately more limited: it is an internal parity check for one configuration, not an estimate of industry-wide implementation risk. The manifest and source fingerprints follow the broader reproducibility objectives of Pineau et al. [2021]. Our contribution lies in connecting those artifacts to explicit economic contracts within one executable evaluation. No individual invariant or accounting identity is claimed as new; the contribution is their executable integration and controlled stress test.

3 Executable audit protocol

We use "contract" in the design-by-contract sense: an executable invariant with an explicit failure response [Meyer, 1992]. Figure 1 summarizes the evaluation path. Each stage can fail independently, and a later stage does not repair an invalid earlier one. Table 1 states the checks used in this study.

3.1 Portfolio, execution, and cash contracts

Let $w_d \in \mathbb{R}^N$ be the risky-asset target decided after observing the close at decision time d, and let $e(d)$ be the first available execution bar strictly after d. At $e(d)$, the primary event engine solves

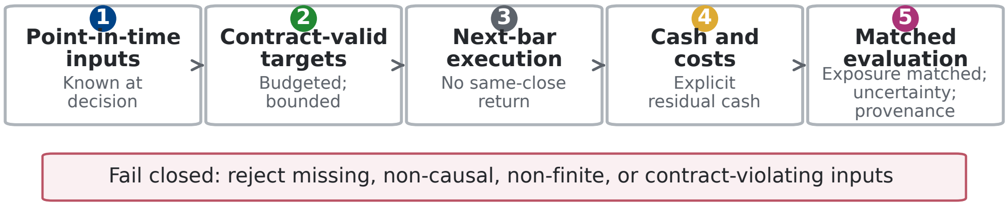

Figure 1: Audit flow from point-in-time inputs to matched evaluation. The audited path validates required inputs, timing, finiteness, and contracts at the stage where each condition becomes testable.

Table 1: Core invariants and failure behavior.

Invariant	Operational check	Failure behavior
Information	Features and fundamentals must be observable by the decision timestamp.	Reject future-dated merges; do not backfill.
Allocation	Allocator output is finite, bounded, and budget-consistent.	Raise a contract error.
Overlay	Timing and risk overlays may reduce exposure but cannot create an invalid book.	Reject non-finite or over-limit weights.
Timing	A close-derived target earns returns only after the first strictly later execution bar.	Shift holdings; never use the observed close twice.
Accounting	Risky assets, residual cash, turnover, and costs share one capital clock.	Reject missing cash bars or cost models.
Comparison	Exact attribution uses realized exposure; a tradable control follows target exposure and pays costs.	Label ex-post controls and cost implementable comparators.
Execution	Persist intent before broker submission; uncertain outcomes are not retried automatically.	Block until broker reconciliation.

adjusted-close pseudo-shares $q_{i,e(d)}$ against post-cost NAV so that the resulting risky allocation matches w_d. It holds q_i fixed between rebalances. If $P_{i,t}$ is the adjusted close and V_t is portfolio NAV, the realized risky weight at the prior close is $h_{i,t-1} = q_{i,t-1}P_{i,t-1}/V_{t-1}$. Close-derived targets therefore earn neither the close they observe nor the return ending at their execution close. For the long-only case, $h_{i,t} \geq 0$ and $\sum_i h_{i,t} \leq 1$; the residual cash holding is

$$h_t^c = 1 - \sum_{i=1}^{N} h_{i,t}. \tag{1}$$

For simple asset returns $r_{i,t}$ over $(t-1, t]$ and cash-account return r_t^c, pre-cost portfolio return is therefore

$$r_t^{\text{gross}} = \sum_{i=1}^{N} h_{i,t-1} r_{i,t} + h_{t-1}^c r_t^c. \tag{2}$$

Missing cash observations fail the run instead of silently becoming zero. For executed risky-notional change $\Delta a_{i,t}$ measured against pre-trade NAV, define $B_t = \sum_i \max(\Delta a_{i,t}, 0)$ and $S_t = \sum_i \max(-\Delta a_{i,t}, 0)$. One-way turnover is $\max(B_t, S_t)$, gross traded notional is $B_t + S_t$, and the charge as a fraction of pre-trade NAV is $\kappa_t = 0.0013(B_t + S_t)$. If V_t^- is pre-trade NAV and V_{t-1} is prior-close NAV, the return drag is $d_t = \kappa_t V_t^- / V_{t-1}$ and $r_t^{\text{net}} = r_t^{\text{gross}} - d_t$. Target shares are solved against post-cost NAV. It does not separately estimate commissions, bid–ask spread, taxes, or market impact. This proportional model is a first-order sensitivity assumption; large or urgent orders require models that account for order size, volatility, and execution urgency, such as Almgren and Chriss [2001].

We report nominal CAGR, maximum drawdown, annualized one-way turnover, annualized gross traded notional, and the conventional sample cash-excess Sharpe statistic

$$\mathrm{SR}_c = \sqrt{252}\,\frac{\mathrm{mean}(r_t - r_t^c)}{\mathrm{sd}(r_t - r_t^c)}. \tag{3}$$

Equation 3 is a descriptive statistic, not an iid-normal inference procedure; serial dependence is handled separately through circular-block resampling [Lo, 2002]. An SHV adjusted-close series supplies the return credited to residual cash; the engine does not trade SHV when cash weight changes. It is a short-duration cash proxy, not a frictionless risk-free asset. Because the input contains adjusted closes, the pseudo-shares are total-return accounting units with implicit distribution and corporate-action adjustment, not broker-faithful nominal shares.

3.2 Exact return attribution

Headline performance does not identify which decision created or destroyed value. Let $x_t = \sum_i h_{i,t-1}$ be risky exposure during return period t, let b_t be the return of an equal-initial-weight buy-and-hold basket, and let $\bar{x}$ be the full-sample mean of x_t. Define the daily exposure-matched passive return and the constant-exposure diagnostic as

$$m_t = x_t b_t + (1 - x_t) r_t^c, \tag{4}$$

$$k_t = \bar{x} b_t + (1 - \bar{x}) r_t^c. \tag{5}$$

Then net excess over cash has the exact arithmetic decomposition

$$\begin{aligned} r_t^{\mathrm{net}} - r_t^c = {} & \underbrace{(r_t^{\mathrm{gross}} - m_t)}_{\text{active risky allocation}} + \underbrace{(m_t - k_t)}_{\text{dynamic exposure timing}} \\ & + \underbrace{(k_t - r_t^c)}_{\text{passive risky exposure}} + \underbrace{(r_t^{\mathrm{net}} - r_t^{\mathrm{gross}})}_{\text{implementation cost}}. \end{aligned} \tag{6}$$

The first term includes group selection, within-risky-asset weighting, and any return effect of the engine's between-rebalance weight semantics; it is not pure security selection in the Brinson sense. The timing term measures daily exposure changes relative to the sample-average exposure. Because $\bar{x}$ is estimated over the full evaluation window, k_t is an ex-post diagnostic rather than a tradable ex-ante benchmark. Equation 6 holds on every date, so arithmetic annualized means add exactly. CAGR does not admit the same additive interpretation.

We resample all five series in Equation 6 with common circular-block indices. Consequently, the identity also holds within every bootstrap draw. The reported fraction of positive draws is a resampling diagnostic, not a posterior probability or a multiple-testing-adjusted p-value.

3.3 Single-assumption diagnostics

Three diagnostics change one assumption while holding the targets, primary engine, and input matrix fixed: residual cash earns zero, proportional costs are removed, or a close-derived target is deliberately executed one bar before that close so it earns the return ending at the observed close. The last construction uses future information and is therefore invalid. Under explicit share holdings, changing the cash return can also change realized exposure between rebalances through the NAV denominator. These are sensitivity checks, not candidate strategies, and they play no role in model selection. The paired design resembles the one-switch logic of Zhang et al. [2026], but here the switches audit cash, costs, and execution semantics in one pipeline.

3.4 Reference implementation and order-state controls

The reference implementation separates data, alpha, portfolio, timing, risk, backtest, execution, and strategy layers. Allocator outputs satisfy a strict budget contract. Post-overlay outputs use a relaxed exposure contract because a timing or risk gate may intentionally leave capital in cash. Point-in-time merge guards reject fundamentals that were not yet public, and optional broker, machine-learning, and provider SDKs remain lazy imports so the scientific core can be tested offline.

A central manifest records the clean source revision, Python and package versions, dependency lock, input SHA-256, experiment design, source-tree hashes, and the hash of every published table

Table 2: Experiment specification recorded in the manifest.

Component	Fixed value
Universe	QQQ, VGT, GLD, TLT, SPY, VIG; QQQ benchmark
Rebalance	First observed session of each calendar week
Group ranking	126-session active-return information ratio; top 2 groups
Residual momentum	126-session estimation + 126-session formation; 21-session gap
Allocation	Positive scores, proportional long-only weights, 60% asset cap
Regime overlay	Rolling vol feature, up to 20 sessions; seeded 3-component full-covariance GMM; 100 EM iterations; 10^{-5} regularization
Volatility overlay	20-point target; 0.3–1.0 multiplier; 21-session proxy
Execution	Event-driven adjusted-close pseudo-shares; first bar strictly after the close-derived decision
Cash and costs	SHV return credited to residual cash; 13 bps of pre-trade gross traded notional
Evaluation	2018-01-02 to 2025-12-31 after 503 warm-up sessions (274 required)
Uncertainty	5,000 joint circular-block draws; seed 42

and figure. A versioned JSON contract and canonical long-form target-tape schema make the engine boundary machine readable. The fixed experiment round-trips every strategy decision stream through the versioned payload boundary and records a canonical payload hash; deterministic fixtures test the same boundary without using the empirical price matrix. The empirical sections below exercise the research and backtest contracts. Order-state controls are verified by deterministic tests: paper-trading state is written atomically as a versioned snapshot, intent is persisted before submission, and an uncertain submission is not retried automatically. SDK-conformance tests instantiate the pinned Alpaca and Interactive Brokers request types. They do not test authenticated transport or venue behavior. Appendix E separates exact identities, property tests, counterfactuals, fault injection, SDK conformance, and untested live behavior.

4 Case study and evaluation design

4.1 Fixed audit configuration

The universe contains QQQ, VGT, GLD, TLT, SPY, and VIG, grouped as growth equities (QQQ, VGT), gold and nominal duration (GLD, TLT), and broad and dividend equities (SPY, VIG). On the first session of each week, the strategy ranks groups by a 126-session information ratio relative to QQQ and keeps the top two. Within selected groups, it estimates a one-factor regression on a preceding 126-session window and sums residual returns over a separate 126-session formation window, skipping the latest 21 sessions. Positive scores receive proportional long-only weights, capped at 60% per asset; if no selected member has a positive score, the target is all cash. QQQ is also one of the two growth-group members, so that group's active-return spread contains one zero benchmark leg. This asymmetry is a fixed design choice.

Two overlays control risky exposure. A seeded three-component Gaussian mixture maps the latest observation to zero, half, or full exposure. Its inputs are QQQ return and rolling realized volatility configured for 20 sessions, represented both as a decimal and in volatility points. The earliest 19 feature rows use the shorter history then available. The two volatility columns are exact rescalings, not independent signals. At each weekly decision, the model is refit on the expanding history available through that close after at least 60 QQQ return observations. Components are ordered by fitted-sample mean return, and only the latest label controls exposure. A separate scaler targets 20 volatility points using a 21-session QQQ realized-volatility proxy because no VIX series is supplied.

The full strategy uses both overlays. Three named variants remove the regime gate, the volatility scaler, or both. Every experiment-facing parameter is recorded in the manifest, and the source-tree digest detects changes to implementation constants. The configuration was frozen for the reported run, but the project was not preregistered and its earlier research trial count is unknown. We therefore report all named variants and make no multiple-testing-adjusted discovery claim.

4.2 Data, comparators, and uncertainty

The owner-supplied matrix is documented as adjusted closes from 2016-01-04 through 2025-12-31, retrieved on 2026-06-16 through yfinance 1.4.1 with `auto_adjust=False`. The yfinance documentation describes the Yahoo Finance API as intended for personal use and directs users to Yahoo's terms of use.[1] The owner authorizes publication of the derived aggregates, but the raw matrix is not distributed. The audited loader requires complete rows and performs no forward filling. The manifest records the provider, retrieval date, adjustment setting, and input SHA-256:

`efb384a62157e56a0cd8065abf45c1ed07d90ec26c681e5d54d74fe4cb9c55e1`

The first 503 sessions are signal history; evaluation covers 2,011 sessions from 2018-01-02 through 2025-12-31.

Comparators share one capital clock. SPY and a six-ETF buy-and-hold basket initialized at equal dollar weights are measured from the close immediately preceding the evaluation window. The exact attribution control multiplies the basket return by the strategy's realized risky exposure each day and assigns the remainder to SHV. It is ex-post and gross of comparator costs, so it is used to explain return rather than presented as tradable. Gross-to-gross and net-to-gross differences against this control remain useful attribution quantities.

We separately run a causal, costed comparator. At each original strategy decision time it targets the strategy's declared gross risky exposure in the same equal-initial-weight buy-and-hold basket, executes on the next bar, assigns residual capital to SHV, and pays the same 13-bps proportional cost. Its initial position is established on the pre-evaluation close, outside the reported window. For Equation 6, the basket is also held at the strategy's full-sample mean exposure. That constant-exposure series is another explicitly ex-post diagnostic.

For dependent daily observations, we jointly resample return-component rows with the circular-block bootstrap of Politis and Romano [1992]. We use 5,000 replications, a 21-session primary block (approximately one trading month), seed 42, and unstudentized two-sided 95% percentile intervals. We repeat the active-return and conventional sample Sharpe calculations with 5- and 63-session blocks to span roughly one week through one quarter. Sharpe is recomputed inside every draw rather than assigned an iid standard error. Intervals quantify sampling variation conditional on this historical path, strategy specification, cash proxy, and block length. They do not establish stationarity or future performance. Fractions of positive draws are descriptive and are not interpreted as significance tests.

5 Results

5.1 Cash accounting changes performance, not the verdict

Table 3 and Figure 2 separate cash and cost effects. Crediting residual cash raises net CAGR from -0.17% under a zero-return cash assumption to 1.40%. That correction is economically material because mean risky exposure is only 30.14% and the strategy is fully in cash on 47.94% of sessions; the exposure path in Figure 2 makes that changing capital allocation explicit. It does not create evidence of alpha. SHV alone compounds at 2.50%. The realized-exposure attribution control compounds at 6.74% before comparator costs; the causal target-exposure comparator compounds at 5.72% after its own modeled costs.

5.2 Allocation and cost explain the shortfall

Equation 6 changes the interpretation of the headline return. Passive risky exposure contributes 3.90% per year; active allocation contributes -3.38%, exposure timing 0.30%, and modeled cost -1.72%. These components sum exactly to -0.90% annualized excess over SHV before rounding.

Table 4 shows that active allocation is the clearest source of underperformance: its 95% interval is [-5.85%, -0.90%], wholly below zero. The timing interval spans zero, so this sample does not isolate a reliable timing effect. Passive exposure is positive, while modeled cost is negative by construction.

[1] https://ranaroussi.github.io/yfinance/

Table 3: Audited accounting over 2018–2025. Conventional sample Sharpe statistics use SHV except for the explicitly zero-cash row, which uses a zero reference rate.

Series	CAGR	Sharpe	Max drawdown
Strategy, after costs and SHV cash	1.40%	-0.15	-9.83%
Strategy, before costs and SHV cash	3.17%	0.14	-8.60%
SHV cash proxy	2.50%	–	-0.45%
Realized-exposure attribution control, gross	6.74%	0.80	-4.64%
Target-exposure comparator, after costs	5.72%	0.62	-4.94%
Strategy, after costs and zero cash	-0.17%	0.00	-15.15%

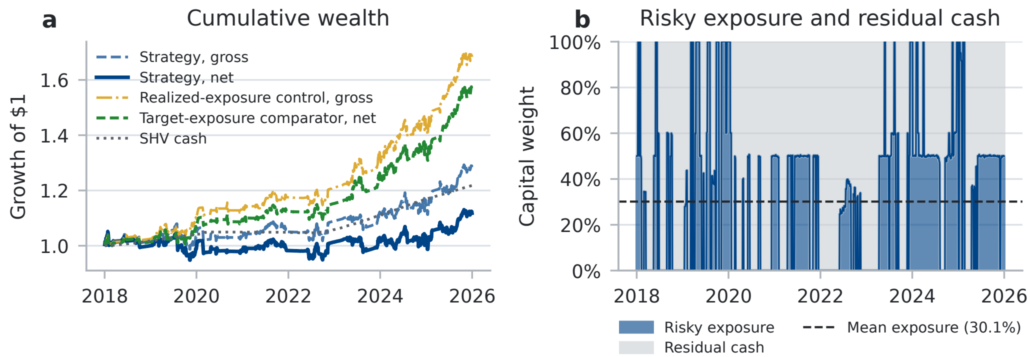

Figure 2: (a) Growth of one dollar under audited cash and cost accounting. (b) The daily capital split between risky assets and residual cash; the dashed line is mean risky exposure. Panel (a) distinguishes the gross ex-post attribution control from the causal target-exposure comparator after costs.

The net-cash interval also spans zero; that does not contradict the stronger matched-comparator result, because the positive passive-exposure component partially offsets allocation and cost losses.

Panel (b) of Figure 3 reports the single-assumption diagnostics. Omitting cash return lowers the CAGR to -0.17%; evaluated relative to SHV, its Sharpe is -0.42. Omitting modeled costs raises the CAGR and Sharpe to 3.17% and 0.14. The deliberately invalid same-close calculation reports 2.49% and 0.03. Thus cost omission and the invalid same-close assignment each change the Sharpe sign relative to the audited value of -0.15. This sign flip is evidence of protocol sensitivity, not evidence that either shortcut is tradable.

5.3 Cost sensitivity and overlay ablations

The strategy turns over 10.30 times per year on a one-way basis and trades 13.25 times NAV per year on a gross two-sided basis. At zero cost, its cash-excess Sharpe is 0.14; it falls to 0.03 at 5 bps, -0.15 at the 13-bps baseline, -0.42 at 25 bps, and -0.96 at 50 bps. The performance estimate is therefore highly sensitive to modest proportional friction.

Appendix Figure 6 compares each variant before costs with a passive basket at the same daily exposure. This avoids attributing the result solely to strategy turnover. The full model has gross cash-excess Sharpe 0.14 versus 0.80 for its comparator. Removing the regime gate, the volatility scaler, or both increases strategy exposure and raw returns, but every gross strategy still trails its matched comparator. The corresponding annualized gross active returns are -3.38%, -5.49%, -3.71%, and -5.54%; every 21-session block-bootstrap interval remains below zero. No variant is selected as a replacement model.

5.4 Uncertainty, subperiods, and engine parity

The net strategy's annualized arithmetic excess return over SHV is -0.90%, with a 95% block-bootstrap interval of [-4.77%, 3.14%]. Before costs, the corresponding estimate is 0.82% [-3.02%,

Table 4: Exact attribution of annualized arithmetic net excess over SHV. Intervals use common 21-session circular-block draws. "Positive" is the number of resampled annualized means above zero, not a posterior probability.

Component	Mean	95% interval	Positive
Active risky allocation	-3.38%	[-5.85%, -0.90%]	15/5000
Dynamic exposure timing	0.30%	[-2.57%, 3.26%]	2889/5000
Passive risky exposure	3.90%	[0.82%, 6.85%]	4963/5000
Modeled implementation cost	-1.72%	[-2.08%, -1.38%]	0/5000
Net excess over cash	-0.90%	[-4.77%, 3.14%]	1682/5000

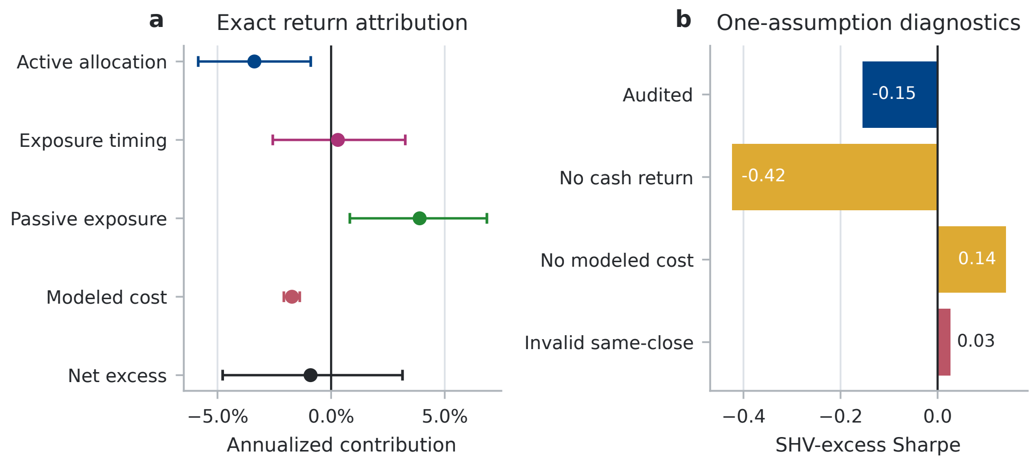

Figure 3: (a) Exact arithmetic return attribution with primary block-bootstrap intervals. (b) One-assumption diagnostics. Amber bars change economic assumptions; the red same-close bar violates causality. None is a strategy alternative.

4.80%]. Neither supports a positive cash-relative claim. Its conventional sample Sharpe is -0.15, with a direct block-bootstrap interval of [-0.80, 0.56].

Relative to the causal comparator after both sides pay modeled costs, annualized arithmetic active return is -4.14% [-6.70%, -1.63%], and conventional active Sharpe is -0.92 [-1.47, -0.37]. Relative to the realized-exposure attribution control, gross active return is -3.38% [-5.85%, -0.90%], and only 0.30% of primary bootstrap draws are positive. After costs it is -5.10% [-7.72%, -2.54%], with no positive primary draw among 5,000.

Appendix Figure 4 shows that changing the circular-block length from 5 to 21 or 63 sessions does not change the matched-comparator conclusion: all before- and after-cost intervals remain below zero.

The costed target-exposure comparator also outperforms in both fixed calendar halves (`subperiods.csv`). The adjusted-close pseudo-share event engine is primary. Its net CAGR and Sharpe are 1.405% and -0.1549, versus 1.404% and -0.1547 for the vectorized diagnostic. This close agreement is local and does not make their rebalancing semantics equivalent.

6 Limitations and broader impact

6.1 Statistical validity

The configuration was frozen only before the reported run. The project was not preregistered, and no complete log of earlier strategy trials survives. Reporting all four named variants prevents selection within that set but cannot recover unknown prior trials; a Deflated Sharpe Ratio or Reality Check would therefore create false precision [White, 2000, Bailey and López de Prado, 2014]. The block bootstrap conditions on one path and approximate dependence through the chosen block length.

Its positive-draw fractions are descriptive, and its marginal intervals are not multiplicity-adjusted. Stability across 5, 21, and 63 sessions is sensitivity evidence, not proof of stationarity or future performance. Component interval endpoints need not add, although the decomposition holds within every draw. The conventional $\sqrt{252}$ Sharpe remains a descriptive statistic; block resampling addresses uncertainty but does not remove the weak-dependence and stationarity assumptions needed to interpret that uncertainty.

6.2 Economic construct validity

"Active risky allocation" combines group selection, within-risky weighting, and the engine's between-rebalance semantics. "Dynamic exposure timing" is relative to full-sample mean exposure, hence diagnostic rather than tradable. The realized-exposure attribution control is not tradable and excludes its own costs. The target-exposure comparator is causal and costed, but its risky leg is a constructed buy-and-hold basket rather than a quoted security; its cost model still omits spread and impact. Arithmetic means decompose exactly; compounded wealth does not.

SHV is only a cash proxy: it has duration, expenses, and market-price variation. Adjusted-close pseudo-shares embed total-return adjustments and omit broker share records, spreads, queue position, and fill uncertainty. The proportional cost grid has no order-size, volatility, venue, impact, ADV, or capacity term. Without VIX, the regime model receives realized volatility twice in different units; its covariance regularization is scale-sensitive, and the separate volatility scaler uses the same underlying proxy.

6.3 External validity and reproducibility

The study covers one U.S.-listed ETF universe, one strategy family, and one eight-year path; it does not estimate failure magnitudes for other settings or match broader multi-model and multi-engine studies [Zhang et al., 2026, Yin et al., 2026]. The provider can revise adjusted histories. The raw input is not redistributed, and provider permission for the derived publication has not been independently verified. Open schemas and conformance fixtures reproduce software semantics, not these returns; an open performance evaluation requires a permitted, reviewer-accessible panel.

6.4 Operational scope and broader impact

Fault-injection tests cover stale or uncertain execution state, but no live authentication, venue behavior, funded order, capacity, or production control was tested. The intended benefit is reduced false confidence; the principal risk is mistaking an audited simulation for investment advice or deployment approval. Paper-mode defaults and fail-closed retries mitigate but do not remove that risk. Past simulated performance does not imply future returns.

7 Conclusion

A target-weight generator does not determine a realized return; information timing, execution, cash, costs, and comparator exposure complete the experiment. Here, crediting residual cash corrects performance but does not create alpha: passive exposure contributes positively, while active allocation and modeled cost erase that benefit. A causal comparator using the same target exposure and cost model reaches the same negative verdict as the exact ex-post attribution. Cost omission or same-close return assignment changes the Sharpe sign without changing the target generator. The contribution is a localized, testable failure analysis rather than a profitable signal.

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A Reproducibility details

The complete generator is `scripts/run_paper_experiments.py`. The release wrapper checks out the committed source revision in a detached clean worktree, writes experiment outputs outside that worktree, verifies the recorded revision and start-state cleanliness, then builds the paper with Tectonic 0.16.9. The reviewed invocation is:

```
scripts/release_paper_artifacts.sh \
  local_data/published_rotation_prices.csv
```

The input must contain adjusted-close columns QQQ, VGT, GLD, TLT, SPY, VIG, and SHV and match the input digest reported in Section 4 for exact reproduction. Aggregate CSVs, generated LaTeX values, figures, and their hashes are listed in `paper/results/manifest.json`. The manifest also records the clean source commit, dependency lock, canonical configuration hash, thread libraries, contract hash, and every package Python file individually. The aggregate source-tree digest is:

`044c219b82295b187b92a7044757289faf91757d276206fe6abdec5d52252e60`

The public code is available at https://github.com/magnaquant/quantcortex. The reviewed run used Python 3.14.4 on an eight-core Apple M1 with 8 GiB RAM and completed in under 40 seconds on CPU; no GPU was used. Supported CI runs use Python 3.11 through 3.14. The reported experiment consumed less than one CPU-minute; repository-wide testing, packaging, notebook execution, and broker conformance checks were additional engineering validation rather than model search.

B Block-length sensitivity

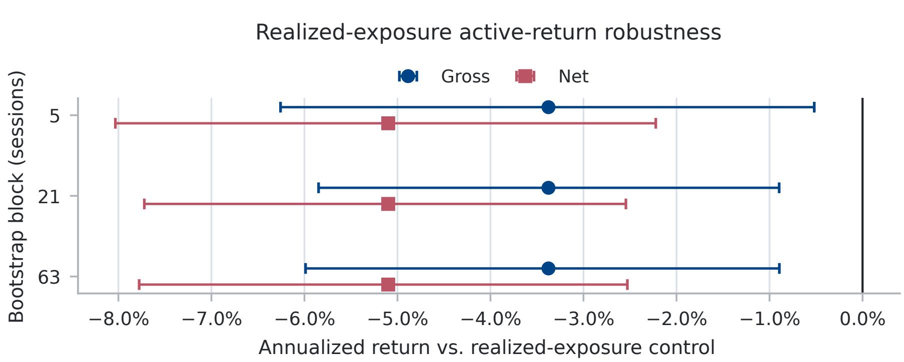

Figure 4: Active-return estimates and 95% percentile intervals under 5-, 21-, and 63-session circular-block resampling against the realized-exposure attribution control. Gross compares before-cost strategy returns with the gross control; net retains modeled strategy costs while the control remains gross.

C Additional engine comparison

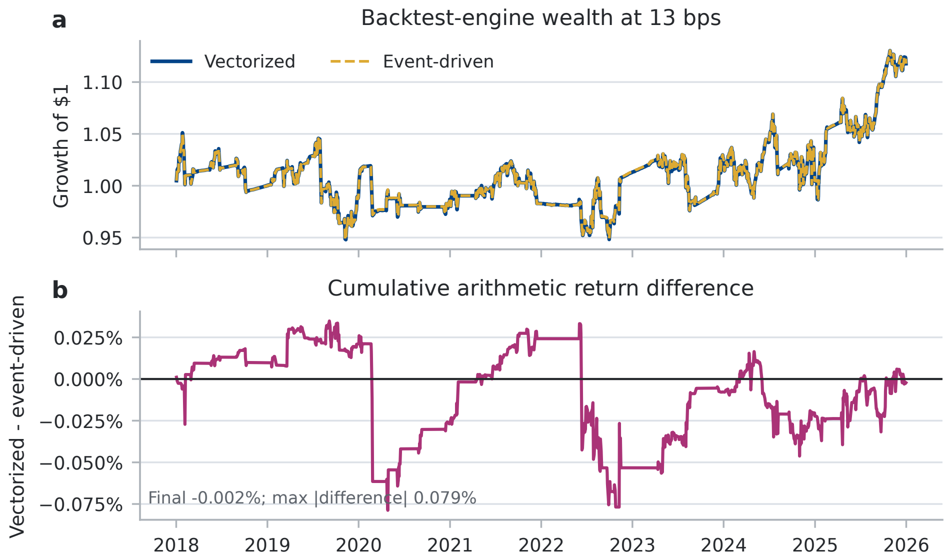

Figure 5: The vectorized engine re-pegs target weights between rebalances, whereas the event-driven engine holds explicit drifting adjusted-close pseudo-shares. Their small difference here does not remove the semantic distinction.

D Cost and overlay sensitivity

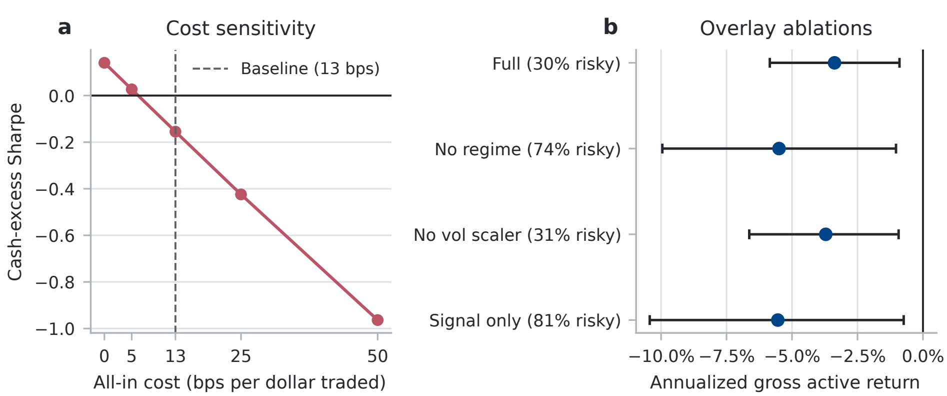

Figure 6: (a) After-cost Sharpe sensitivity for the full strategy. (b) Annualized gross strategy return minus the realized-exposure attribution control, with 95% intervals from common 21-session circular-block draws. Tick labels report mean risky exposure; all named audit variants are shown.

E Contract-to-evidence map

The artifact distinguishes evidence types rather than treating every passing check as equivalent:

Evidence class	Claim supported	Committed evidence
Exact identity	Daily return components reconstruct net cash excess	Per-row assertion and `return_decomposition.csv`
Property test	Weight, timing, cash, and target-tape invariants hold	Deterministic pytest cases, including hand-computable tapes
Counterfactual	One declared assumption changes the reported metric	`protocol_switches.csv`
Fault injection	Unsafe or stale execution state fails closed	Truncated reconciliation, unknown status, and stale-writer tests
SDK conformance	Offline request and response mappings match pinned SDKs	Broker model-construction tests and behavioral mocks
Untested live	Authenticated transport, venue state, and funded orders	No supporting evidence; explicitly outside scope

F Use of language-model assistance

The author used an LLM-based coding assistant for repository inspection, code review, regression-test drafting, and copyediting. The author retained responsibility for the strategy, data, statistical methods, experiment design, and all reported claims, and reviewed the resulting code and prose. Numerical claims are generated by committed scripts and machine-readable artifacts.

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